Entropy and space complexity of random approximate sets

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Abstract

We derive the joint entropy of the false positive and false negative error counts in the Bernoulli set model, a model of random approximate sets with independent, element-wise errors. The joint entropy decomposes additively via element independence and admits a closed-form asymptotic expansion based on the binomial entropy approximation. We then establish the information-theoretic space lower bound for positive approximate sets over countably infinite universes, restating and proving the classical result of Carter, Floyd, Gill, Markowsky, and Wegman in the notation of the Bernoulli framework. The Bloom filter achieves $\ln 2 \approx 69\%$ of the optimal density. For approximate sets with both false positives and false negatives ($0 < \tau < 1$), the corresponding bound of about $\tau \log_2(1/\varepsilon)$ bits per element is due to Pagh and Rodler (2001); an earlier version of this preprint wrongly called this case open.

Status and errata (October 2026). This is the final version of this preprint; no further revisions are planned. It is superseded by Chapter 8 (*Information-Theoretic Bounds*) of the monograph *Bernoulli Types: Two-Sided Error in Approximate Membership, Retrieval, and Computation* (A. Towell, 2026), https://metafunctor.com/latex/bernoulli-monograph/book.html (hub: https://metafunctor.com/research/bernoulli/). Please cite the monograph in preference to this preprint.

Corrections in this version.

- Abstract, Secs. 1, 4, 5: the two-sided bound ($0 < \tau < 1$) is no longer called open; it is Pagh and Rodler [8, Prop. 1].
- Prop. 4.1, proof: the last step is now $B \geq -p \log_2 \varepsilon - o(1)$ as $n \to \infty$, not an exact inequality.
- Prop. 4.1, proof: the randomized case is sketched as the standard averaging argument of Yao [15], not asserted.
- Sec. 4: $\mathcal{O}(u)$ bits for a dense set is an upper bound, not a requirement.
- Sec. 4: the perfect hash filter is credited to Dietzfelbinger and Pagh [5, Sec. 1], matching constructions to [7, 5], seed search to [6].
- Sec. 1: both results are presented as exposition, not as contributions.
- Sec. 5: mutual information and capacity are no longer called unexplored; binary channel capacity is classical [10].
- Bibliography: companion working papers now cite their Zenodo DOIs.

Predecessors credited. One-sided bound: Carter et al. [3]. Two-sided bound: Pagh and Rodler [8]. Matching constructions: Pagh, Pagh, and Rao [7]; Dietzfelbinger and Pagh [5]. Bloom filter cost: [1, 2]. The joint entropy of the error counts follows from element independence and the binomial entropy expansion [4]. No novelty is claimed for these results. Dated changes: `ERRATA.md` in the source repository.

Contents

1 Introduction 2

2 Preliminaries 3

3 Entropy of approximate sets **3**
3.1 Positives and negatives 4

4 Space complexity **4**

5 Conclusion **6**

1 Introduction

The Bernoulli set model [12] provides a compositional framework for random approximate sets—data structures whose membership queries may err with known false positive and false negative rates. A natural information-theoretic question is: how much uncertainty do these errors carry?

Motivation. The joint entropy of error counts quantifies the inherent unpredictability of a Bernoulli set realization. This entropy has direct implications for coding: any lossless representation of the error pattern requires at least this many bits on average. The space complexity question—how many bits are needed to implement an approximate set—connects the abstract Bernoulli model to the concrete cost of probabilistic data structures.

Content. The paper collects two results. The first is an elementary consequence of element independence and the standard binomial entropy expansion; the second is classical. First, we derive the joint entropy of the false positive and false negative counts (theorem 3.1), showing that it decomposes additively by element independence and admits the closed-form asymptotic expansion

$$\log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot\varepsilon(1-\varepsilon)\cdot\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right). \tag{1.1}$$

We extend this to the case where the number of positives is itself uncertain (section 3.1). Second, we restate and prove the classical space lower bound of Carter et al. [3] for positive approximate sets (proposition 4.1): any implementation over a countably infinite universe requires at least $-\log_2\varepsilon$ bits per element. We compare this to the Bloom filter's $-(1/\ln 2)\log_2\varepsilon$ bits per element. The general case $0 < \tau < 1$ is not open: the lossy-dictionary bound of Pagh and Rodler [8] gives about $\tau\log_2(1/\varepsilon)$ bits per element (section 4). An earlier version of this preprint stated otherwise.

Related work. Shannon [9] introduced the entropy of binary channels; Cover and Thomas [4] provide the binomial entropy expansion we use. Carter et al. [3] established the space lower bound for structures without false negatives, and Pagh and Rodler [8] extended it to structures that also drop a fraction of the keys. Pagh, Pagh, and Rao [7] and Dietzfelbinger and Pagh [5] give constructions that meet the one-sided bound up to lower-order terms. The extension of a counting bound from deterministic to randomized structures is the averaging argument of Yao [15]. Bloom [1] introduced the eponymous filter; Broder and Mitzenmacher [2] survey its analysis. The Bernoulli set model and its compositional algebra are developed in a companion paper [12].

Organization. Section 2 restates the Bernoulli axioms. Section 3 derives the joint entropy. Section 4 establishes the space complexity lower bound. Section 5 concludes.

2 Preliminaries

We restate the elements of the Bernoulli set model [12] used in this paper.

Axiom 1 (Element-wise independence)**.** *For all distinct $x, y \in U$, the error events at x and y are mutually independent:*

$$P(\mathbf{1}_{\tilde{A}}(x) \neq \mathbf{1}_A(x) \mid \mathbf{1}_{\tilde{A}}(y) \neq \mathbf{1}_A(y)) = P(\mathbf{1}_{\tilde{A}}(x) \neq \mathbf{1}_A(x)). \tag{2.1}$$

More generally, the family $\{\mathbf{1}_{\tilde{A}}(x) : x \in U\}$ is mutually independent given the objective set A and the error rates. Furthermore, within each partition block B_i, the error indicators $\{\mathbf{1}_{\tilde{A}}(x) : x \in B_i\}$ are identically distributed.

Axiom 2 (Conditional independence of block error rates)**.** *Given the objective set R, the random error rates $\alpha_1, \ldots, \alpha_n$ for the respective partition blocks $B_1, \ldots, B_n$ are mutually independent.*

From these axioms, the false positive count $\mathrm{FP}_m \sim \mathrm{Bin}(u - m, \varepsilon)$ and the false negative count $\mathrm{FN}_m \sim \mathrm{Bin}(m, \omega)$ are conditionally independent given m positives in a universe of u elements [12].

3 Entropy of approximate sets

The Bernoulli set model assigns binomial distributions to the number of false positives and false negatives. A natural question is how much *uncertainty* these error counts carry, as measured by Shannon entropy [9]. In this section, we derive the joint entropy of the false positive and false negative counts and extend the analysis to the case in which the number of positives is itself uncertain.

Theorem 3.1 (Joint entropy of false positives and false negatives)**.** *Given m positives in a universe of u elements, the joint entropy of the uncertain number of false positives and false negatives in a Bernoulli set with expected false positive rate ε and expected false negative rate ω is*

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot \varepsilon(1-\varepsilon)\cdot \omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right), \tag{3.1}$$

where the asymptotic term holds as $u \to \infty$ with m/u bounded away from 0 and 1 (so that both m and $u - m$ grow with u).

Proof. By axiom 1, the random error counts FP_m and FN_m are conditionally independent given m positives (they are functions of indicators on disjoint element sets: negatives and positives respectively). Therefore the joint entropy decomposes additively:

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \mathcal{H}(\mathrm{FP}_m) + \mathcal{H}(\mathrm{FN}_m). \tag{a}$$

The random variable FP_m is binomially distributed, $\mathrm{FP}_m \sim \mathrm{Bin}(u - m, \varepsilon)$. The entropy of a binomial distribution $\mathrm{Bin}(n, p)$ admits the well-known asymptotic expansion [4]

$$H = \frac{1}{2}\log_2\Big(2\pi e \cdot n \cdot p(1-p)\Big) + \mathcal{O}(1/n). \tag{b}$$

Applying eq. (b) to FP_m and FN_m respectively yields

$$\mathcal{H}(\mathrm{FP}_m) = \frac{1}{2}\log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\,\varepsilon(1-\varepsilon)\Big) + \mathcal{O}\left(\frac{1}{u-m}\right), \tag{c}$$

$$\mathcal{H}(\mathrm{FN}_m) = \frac{1}{2}\log_2(2\pi e) + \frac{1}{2}\log_2\Big(m\,\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{1}{m}\right). \tag{d}$$

Summing these expressions and combining the logarithms gives

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\,m\,\varepsilon(1-\varepsilon)\,\omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right), \tag{e}$$

where the error term follows from $\mathcal{O}(1/(u-m)) + \mathcal{O}(1/m) = \mathcal{O}\Big(u/(m(u-m))\Big)$. □

3.1 Positives and negatives

When the number of positives is itself uncertain, modeled by the random variable P with probability mass function $\mathrm{p_P}(p|u)$, the joint distribution of positives, false positives, and false negatives factorizes as

$$\mathrm{p}(p, f_p, f_n|u, \varepsilon, \omega) = \mathrm{p_P}(p|u)\ \mathrm{p_{FP}}(f_p|p, u, \varepsilon)\ \mathrm{p_{FN}}(f_n|p, u, \omega). \tag{3.2}$$

By axiom 1, FP and FN are conditionally independent given P, so the joint entropy decomposes as

$$\mathcal{H}(\mathrm{P}, \mathrm{FP}, \mathrm{FN}) = \mathcal{H}(\mathrm{P}) + \sum_{p=0}^{u} \mathrm{p_P}(p|u)\ \mathcal{H}(\mathrm{FP} \mid p) + \sum_{p=0}^{u} \mathrm{p_P}(p|u)\ \mathcal{H}(\mathrm{FN} \mid p). \tag{3.3}$$

The last two sums are the conditional entropies $\mathcal{H}(\mathrm{FP} \mid \mathrm{P})$ and $\mathcal{H}(\mathrm{FN} \mid \mathrm{P})$, each weighted by the distribution of positives.

Example 1 *The* expected *number of false positives, marginalizing over* P*, is*

$$\begin{aligned}\mathrm{E}\left[\mathrm{FP}\right] &= \sum_{p=0}^{u} \mathrm{p_P}(p|u)\,\mathrm{E}\left[\mathrm{FP}_p\right] = \sum_{p=0}^{u} \mathrm{p_P}(p|u)\,(u-p)\,\varepsilon \\ &= \varepsilon\Big(u - \mathrm{E}\left[\mathrm{P}\right]\Big) = \varepsilon \cdot \mathrm{E}\left[\mathrm{N}\right],\end{aligned} \tag{a}$$

confirming that the marginal expected false positive count depends only on the expected number of negatives and the false positive rate ε.

4 Space complexity

If the finite cardinality of a universe is u and the set is *dense* (and the approximation is also dense, i.e., the false negative rate is relatively small), then

$$\mathcal{O}(u) \text{ bits} \tag{4.1}$$

suffice to code the set (an exact bit vector), which is independent of p, the false positive rate, and the false negative rate.

The lower-bound on the *expected* space complexity of a data structure that models the *positive random approximate set* where the elements are over a *countably infinite* universe is a classical result due to Carter, Floyd, Gill, Markowsky, and Wegman [3]. We restate it here in the notation of the present paper and include a proof for completeness.

Proposition 4.1 ([3])**.** *The* information-theoretic lower-bound *of a data structure that implements the countably infinite* positive random approximate set *abstract data type has an* expected *bit length given by*

$$-\log_2 \varepsilon \text{ bits/element}, \tag{4.2}$$

where $\varepsilon > 0$ is the false positive rate.

Proof. We give the counting argument for deterministic fixed-length encodings. The extension to randomized structures is the standard averaging argument behind Yao's principle [15], sketched here and not claimed as new. Draw S uniformly from the size-p subsets and regard a randomized structure as a distribution over deterministic encodings indexed by its seed R. Given R and the encoded state, S is a size-p subset of the positive response set, so its conditional entropy is at most the expectation of $\log_2 \binom{p+x(n-p)}{p}$, where x is the realized fraction of negatives accepted. That function is concave in x and $\mathrm{E}[x] \le \varepsilon$, so by Jensen's inequality the bound below holds with B read as the expected length of a prefix-free encoding, which is at least the entropy of the encoded state.

Consider a finite subset of the universe with n elements, from which an objective set S of size p is drawn. A deterministic encoding using B bits maps each input set S to a state $r = f(S)$ from at most 2^B distinct states. Each state r determines a *positive response set* $Q(r) = \{x : \mathrm{query}(x, r) = 1\}$. Since the data structure has no false negatives ($\tau = 1$), we have $S \subseteq Q(r)$ for the state $r = f(S)$ encoding S. The false positive rate constraint requires that at most an ε-fraction of the $n - p$ negatives belong to $Q(f(S))$ for every input set S, so $|Q(f(S))| \le p + \varepsilon(n - p)$ for all S. In particular, $|Q(r)| \le p + \varepsilon(n - p)$ for every state r in the image of f.

By the pigeonhole principle, some state r^* must encode at least $\binom{n}{p}/2^B$ distinct input sets. Every such set is a subset of $Q(r^*)$ of size p, so

$$\binom{n}{p}/2^B \le \binom{|Q(r^*)|}{p} \le \binom{p + \varepsilon(n-p)}{p}. \tag{a}$$

Taking logarithms and rearranging:

$$B \ge \log_2 \binom{n}{p} - \log_2 \binom{p + \varepsilon(n-p)}{p}. \tag{b}$$

For fixed p as $n \to \infty$, $\binom{n}{p} \sim n^p/p!$ and $\binom{p+\varepsilon n}{p} \sim (\varepsilon n)^p/p!$, so the right-hand side increases to $\log_2\!\left(n^p/(\varepsilon n)^p\right) = -p \log_2 \varepsilon$ from below and

$$B \ge -p \log_2 \varepsilon - o(1) \qquad (n \to \infty), \tag{c}$$

giving $-\log_2 \varepsilon$ bits per element in the countably infinite limit. This is a fixed-size, growing-universe regime, unlike the constant-density regime of theorem 3.1. □

Remark (The two-sided case)**.** *When false negatives occur ($0 < \tau < 1$), $S \not\subseteq Q(r)$ and the pigeonhole step above does not apply directly. An earlier version of this preprint called this case open; it is not. Pagh and Rodler [8, Prop. 1] show that a lossy dictionary storing n keys from a universe of size 2^w, with at most γn false negatives and false positive rate ε, needs at least $(1-\gamma)\, n \log_2\!\left(1/(\varepsilon + n/2^w)\right) - \mathcal{O}(n)$ bits. In the large-universe limit, with $\gamma = 1 - \tau$, this is $-\tau \log_2 \varepsilon - \mathcal{O}(1)$ bits per element, so the scaling $-\tau \log_2 \varepsilon$ (retaining $p\tau$ elements, each costing $-\log_2 \varepsilon$ bits) is a valid lower bound up to an additive constant per element. Their model bounds the number of false negatives, whereas the Bernoulli model fixes their expected rate; we do not carry out that transfer here.* △

A well-known implementation of countably infinite *positive approximate set* is the Bloom filter [1] which has an expected space complexity given by [2]

$$-\frac{1}{\ln 2} \log_2 \varepsilon \text{ bits/element}, \tag{4.3}$$

thus the absolute efficiency of the Bloom filter is $\ln 2 \approx 0.69$. Constructions that meet the bound of proposition 4.1 up to lower-order terms are known [7, 5]. A practical implementation of the *positive random approximate set* is given by the *Perfect Hash Filter* [14], a minimal perfect hash function that stores a $\log_2(1/\varepsilon)$-bit fingerprint per element, as described by Dietzfelbinger and Pagh [5, Sec. 1]; it compares favorably to the Bloom filter in many circumstances.[1]

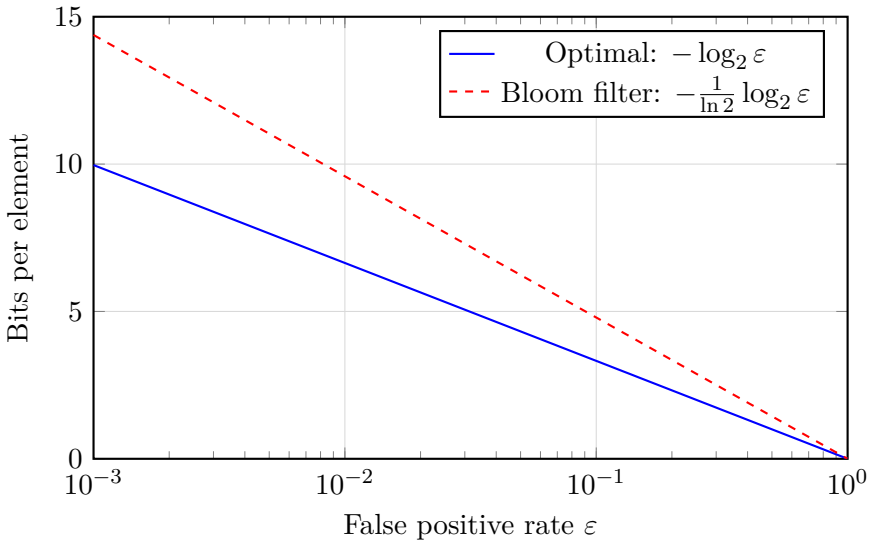


Figure 1: Space complexity of positive approximate sets as a function of the false positive rate ε. The solid curve is the information-theoretic lower bound $-\log_2 \varepsilon$ bits per element (proposition 4.1). The dashed curve is the Bloom filter cost $-(1/\ln 2)\log_2 \varepsilon$; the gap reflects the Bloom filter's $\ln 2 \approx 69\%$ efficiency.

5 Conclusion

We derived two information-theoretic results for the Bernoulli set model. First, the joint entropy of the false positive and false negative counts decomposes additively by element independence (theorem 3.1), yielding the closed-form asymptotic expansion

$$\mathcal{H}(\mathrm{FP}_m, \mathrm{FN}_m) = \log_2(2\pi e) + \frac{1}{2}\log_2\Big((u-m)\cdot m\cdot \varepsilon(1-\varepsilon)\cdot \omega(1-\omega)\Big) + \mathcal{O}\left(\frac{u}{m(u-m)}\right). \quad (5.1)$$

This result extends naturally to the case in which the number of positives is itself uncertain (section 3.1). Second, we restated and proved the classical information-theoretic space lower bound of Carter et al. [3] for positive approximate sets over countably infinite universes (proposition 4.1): at least $-\log_2 \varepsilon$ bits per element are required. The Bloom filter [1] achieves $\ln 2 \approx 69\%$ of this optimal density.

[1] The *Bernoulli hash function* [11] (earlier called the singular hash set) attains the bound by *brute-force* search for a seed, so it is not practical for even relatively small objective sets; its purpose is analytic tractability. Single-seed search of this kind, with storage near the information scale and exponential construction work, is analyzed by Lehmann et al. [6].

The two-sided case. The information-theoretic space bound of proposition 4.1 assumes a positive approximate set ($\tau = 1$). The general case $0 < \tau < 1$ is not open, contrary to an earlier version of this preprint: Pagh and Rodler [8] give a lower bound of $-\tau \log_2 \varepsilon - \mathcal{O}(1)$ bits per element in the large-universe limit for structures with at most a $1 - \tau$ fraction of false negatives (section 4).

Further directions. The mutual information $I(\tilde{A}; A)$ between a Bernoulli set and its objective set is not treated here. Under element-wise independence and an independent prior on membership, it reduces to a sum of per-element quantities of a binary asymmetric channel, whose capacity is classical [10].

Extending the entropy analysis to compositions of Bernoulli sets—where error rates are themselves functions of operand rates—would connect the present results to the compositional algebra of [13].

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